Stochastic Processes Semester 1

Tutorial 8 – Solutions

1. (a) First we check the given p_{ij}s satisfy the initial conditions:

$$p_{ij}(0) = \begin{cases} \frac{1}{3} + \frac{2}{3}\exp(0) = 1 & \text{if } j = i \\ \frac{1}{3} - \frac{1}{3}\exp(0) = 0 & \text{if } j \neq i. \end{cases}$$

Next we compute their derivatives with respect to t:

$$p'_{ij}(t) = \begin{cases} -2\lambda\exp(-3\lambda t) & \text{if } j = i \\ \lambda\exp(-3\lambda t) & \text{if } j \neq i. \end{cases}$$

Now we check that the FDEs

$$p'_{ij}(t) = \sum_{k=1}^{3} p_{ik}(t)q_{kj} \quad \text{for all } i, j \in S = \{1, 2, 3\}$$

are satisfied by $p_{ij}(\cdot)$ as given. There are 2 cases: $i = j$ and $i \neq j$. So for $i = j$ we have

$$\begin{aligned} \sum_{k=1}^{3} p_{ik}(t)q_{ki} &= p_{ii}(t)q_{ii} + 2p_{ik}(t)q_{ki} \\ &= \left(\frac{1}{3} + \frac{2}{3}\exp(-3\lambda t)\right)(-2\lambda) + 2\left(\frac{1}{3} - \frac{1}{3}\exp(-3\lambda t)\right)(\lambda) \\ &= -2\lambda\exp(-3\lambda t) = p'_{ii}(t) \end{aligned}$$

and for $i \neq j$ we have

$$\begin{aligned} \sum_{k=1}^{3} p_{ik}(t)q_{kj} &= p_{ii}(t)q_{ij} + p_{ij}(t)q_{jj} + p_{ik}(t)q_{kj} \\ &= \lambda\left(\frac{1}{3} + \frac{2}{3}\exp(-3\lambda t)\right) - 2\lambda\left(\frac{1}{3} - \frac{1}{3}\exp(-3\lambda t)\right) + \lambda\left(\frac{1}{3} - \frac{1}{3}\exp(-3\lambda t)\right) \\ &= \lambda\exp(-3\lambda t) = p'_{ij}(t) \end{aligned}$$

(b) We can find these p_{ij}s directly as follows. Firstly, by the symmetry of the matrix we know that $p_{ij}(t) = p_{ik}(t)$ for $j, k \neq i$ and that $p_{ii}(t) = p_{jj}(t)$ for all i, j. Also $p_{ii}(t) + 2p_{ij}(t) = 1$ using the fact that the sum of every row of $P(t)$ is 1.
Let us consider the FDE for $p'_{ii}(t)$:

$$\begin{aligned} p'_{ii}(t) &= p_{ii}(t)q_{ii} + 2p_{ik}(t)q_{ki} \\ &= p_{ii}(-2\lambda) + (1 - p_{ii}(t))(\lambda) = \lambda - 3\lambda p_{ii}(t). \end{aligned}$$

Re-arranging, this gives:

$$p'_{ii}(t) + 3\lambda p_{ii}(t) = \lambda.$$

So we can use the integrating factor method with

$$I(t) = \exp(3\lambda t)$$

to give

$$\exp(3\lambda t)p_{ii}(t) = \int \lambda\exp(3\lambda t)dt$$

and so

$$p_{ii}(t) = 1/3 + c\exp(-3\lambda t).$$

Now use the fact that $p_{ii}(0) = 1$ to get $c = 2/3$ and so

$$p_{ii}(t) = 1/3 + 2/3\exp(-3\lambda t).$$

For $i \neq j$, we have

$$p'_{ij}(t) = \lambda p_{ii}(t) - \lambda p_{ij}(t) = \lambda - 3\lambda p_{ij}(t),$$

which is the same equation as for $p'_{ii}(t)$, so we have

$$p_{ij}(t) = 1/3 + k\exp(-3\lambda t).$$

Now using the initial condition $p_{ij}(0) = 0$ we see that $k = -1/3$ giving the required equation.

(c) We solve $\pi Q = 0$:

$$\begin{aligned} 0 &= -2\lambda\pi_1 + \lambda\pi_2 + \lambda\pi_3 \\ 0 &= \lambda\pi_1 + -2\lambda\pi_2 + \lambda\pi_3 \\ 0 &= \lambda\pi_1 + \lambda\pi_2 + -2\lambda\pi_3 \\ 1 &= \pi_1 + \pi_2 + \pi_3 \end{aligned}$$

This gives $\boldsymbol{\pi} = (1/3, 1/3, 1/3)$. This makes sense because of the symmetry of the matrix.

(d) We have

$$\lim_{t\to\infty} p_{ij}(t) = \lim_{t\to\infty} \begin{cases} \frac{1}{3} + \frac{2}{3}\exp(-3\lambda t) & \text{if } j = i \\ \frac{1}{3} - \frac{1}{3}\exp(-3\lambda t) & \text{if } j \neq i. \end{cases}.$$

Both of these limits are 1/3.

2. (a) First, it is straightforward to see that the given equations satisfy satisfies the initial conditions $p_{HH}(0) = 1$, $p_{HS}(0) = p_{HD}(0) = 0$.
The forward differential equations associated with the initial state H are

$$\begin{aligned} p'_{HH}(t) &= -(\lambda + \nu)p_{HH}(t) + \mu p_{HS}(t) \\ p'_{HS}(t) &= \lambda p_{HH}(t) - (\mu + \nu)p_{HS}(t) \\ p'_{HD}(t) &= \nu(p_{HH}(t) + p_{HS}(t)) \end{aligned}$$

Let us check that the given solution (i) satisfies these differential equations. In fact, it is sufficient to check that the first two of the above differential equations are satisfied and that $p_{HH}(t) + p_{HS}(t) + p_{HD}(t) = 1$ for all t.
Differentiating $p_{HH}(t)$, $p_{HS}(t)$ and $p_{HD}(t)$ we have

$$\begin{aligned} p'_{HH}(t) &= \left(\frac{-\nu\mu}{\lambda+\mu} + \frac{-(\lambda+\mu+\nu)\lambda}{\lambda+\mu}e^{-(\lambda+\mu)t}\right)e^{-\nu t} \\ p'_{HS}(t) &= \left(\frac{-\nu\lambda}{\lambda+\mu} - \frac{-(\lambda+\mu+\nu)\lambda}{\lambda+\mu}e^{-(\lambda+\mu)t}\right)e^{-\nu t} \\ p'_{HD}(t) &= \nu e^{-\nu t} \end{aligned}$$

from which we can see that these derivatives satisfy the above FDEs.

(b) For the remaining probability functions, we have

$$\begin{aligned} p_{SH}(t) &= \left(\frac{\mu}{\lambda+\mu} - \frac{\mu}{\lambda+\mu}e^{-(\lambda+\mu)t}\right)e^{-\nu t} \\ p_{SS}(t) &= \left(\frac{\lambda}{\lambda+\mu} + \frac{\mu}{\lambda+\mu}e^{-(\lambda+\mu)t}\right)e^{-\nu t} \\ p_{HD}(t) &= 1 - e^{-\nu t} \end{aligned}$$

and

$$p_{DH}(t) = 0, \qquad p_{DS}(t) = 0, \qquad p_{DD}(t) = 1,$$

since D is an absorbing state.

(c) First, noticing that $q_{SD} = q_{HD} = \nu$ gives the FDE:

$$p_{HD}^{'}(t) = \nu(p_{HH}(t) + p_{HS}(t)) = \nu(1 - p_{HD}(t)).$$

Rearranging we have $p_{HD}^{'}(t) + \nu p_{HD}(t) = \nu$ and so using the integrating factor method we have $I(t) = \exp(\nu t)$ giving

$$\exp(\nu t)p_{HD}(t) = \int \nu \exp(\nu t)dt = \exp(\nu t) + c.$$

Using the condition $p_{HD}(0) = 0$ gives $c = -1$ and so

$$p_{HD}(t) = 1 - e^{-\nu t}.$$

We now move onto the FDE for $p_{HH}^{'}(t)$:

$$p_{HH}^{'}(t) = p_{HH}(t)(-\lambda - \nu) + p_{HS}(t)\mu.$$

Since $p_{HH}(t) + p_{HS}(t) + p_{HD}(t) = 1$, we get

$$p_{HH}^{'}(t) = p_{HH}(t)(-\lambda - \nu) + (1 - p_{HH}(t) - 1 + e^{-\nu t})\mu$$

which gives

$$p_{HH}^{'}(t) + p_{HH}(t)(\lambda + \mu + \nu) = e^{-\nu t}\mu.$$

Multiplying both sides by $e^{(\lambda+\mu+\nu)t}$, we get

$$\left(p_{HH}(t)e^{(\lambda+\mu+\nu)t}\right)^{'} = \mu e^{(\lambda+\mu)t}.$$

Then the result follows by integrating and using the boundary condition at $t = 0$.

(d) Either directly or by noting that D is an absorbing state,

$$\lim_{t\to\infty} p_{ij}(t) = \begin{cases} 0 & \text{if } j = H \text{ or } j = S \\ 1 & \text{if } j = D. \end{cases}$$

(e) The transition matrix of the associated jumping chain is

$$P^* = \begin{pmatrix} 0 & \frac{\lambda}{\lambda+\nu} & \frac{\nu}{\lambda+\nu} \\ \frac{\mu}{\mu+\nu} & 0 & \frac{\nu}{\mu+\nu} \\ 0 & 0 & 1 \end{pmatrix},$$

from which it is again observed D is an absorbing state while H and S are transient. Hence eventual absorption in state D is certain.

3. (a) The transition matrix of the corresponding jumping chain is given by

$$P^* = \begin{pmatrix} 0 & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & 0 & \frac{1}{4} & \frac{3}{4} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

(b) From the transition matrix of the corresponding jumping chain we see that the classes are:

$\{1,2,3\}$ open, hence transient;
$\{4\}$ closed, hence (since also finite) recurrent;
$\{5,6\}$ closed, hence (since also finite) recurrent.

Note that state 4 is in fact a single absorbing state.

(c) For each state i, let $\alpha_i^{\{4\}} = \mathbf{P}(\text{process } X(\cdot) \text{ absorbed in state } 4 | X(0) = i)$. Then, solving for these absorption probabilities via the equation $\boldsymbol{\alpha}^{\{4\}} = \boldsymbol{\alpha}^{\{4\}} P^*$, we have

$$\boldsymbol{\alpha}^{\{4\}} = \left(\frac{19}{35}, \frac{33}{35}, \frac{27}{35}, 1, 0, 0 \right).$$

Again arguing as we did in Chapter 4, we have

$$\alpha_i^{\{5,6\}} = \mathbf{P}(\text{process } X(\cdot) \text{ absorbed in class } \{5,6\} | X(0) = i) = 1 - \alpha_i^{\{4\}},$$

since there are only a finite number of states. It follows that

$$\boldsymbol{\alpha}^{\{5,6\}} = \left(\frac{16}{35}, \frac{2}{35}, \frac{8}{35}, 0, 1, 1 \right).$$

4. We solve the equations $\boldsymbol{\pi} Q = \mathbf{0}$:

$$\begin{aligned} -3\pi_1 + 3\pi_2 + 4\pi_3 &= 0 \\ \pi_1 - 4\pi_2 + 2\pi_3 &= 0 \\ 2\pi_1 + \pi_2 - 6\pi_3 &= 0 \end{aligned}$$

with the additional condition $\pi_1 + \pi_2 + \pi_3 = 1$ to obtain the stationary distribution $\boldsymbol{\pi} = (22/41, 10/41, 9/41)$.

5. (a) i. The equation $\boldsymbol{\pi} Q = \mathbf{0}$ here becomes

$$\begin{aligned} -\pi_0 \lambda + \pi_1 \mu &= 0 \\ \pi_{i-1} \lambda - \pi_i (\lambda + \mu) + \pi_{i+1} \mu &= 0, \qquad i \geq 1. \end{aligned}$$

We now subsitute $\pi_i = k \left(\frac{\lambda}{\mu} \right)^i$ into the left hand sides of both equations:

$$-k \left(\frac{\lambda}{\mu} \right)^0 \lambda + k \left(\frac{\lambda}{\mu} \right)^1 \mu = -k + k = 0$$

$$k \left(\frac{\lambda}{\mu} \right)^{i-1} \lambda - k \left(\frac{\lambda}{\mu} \right)^i (\lambda + \mu) + k \left(\frac{\lambda}{\mu} \right)^{i+1} \mu = k \left(\frac{\lambda^i \mu - \lambda^{i+1} - \lambda^i \mu + \lambda^{i+1}}{\mu^i} \right) = 0.$$

ii. For a stationary distribution $\boldsymbol{\pi}$, we require

$$\sum_{i \geq 0} \pi_i = \sum_{i \geq 0} k \left(\frac{\lambda}{\mu} \right)^i = k \sum_{i \geq 0} \left(\frac{\lambda}{\mu} \right)^i = 1.$$

The infinite geometric sum

$$\sum_{i\geq 0}\left(\frac{\lambda}{\mu}\right)^i$$

converges if and only if $\lambda < \mu$ (since we know $\lambda, \mu > 0$). For $\lambda < \mu$,

$$\sum_{i\geq 0}\left(\frac{\lambda}{\mu}\right)^i = \frac{1}{1-\lambda/\mu}$$

and so $k = 1 - \lambda/\mu$. Thus

$$\pi_i = \left(1-\frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^i, \qquad i \geq 0.$$

(b) For the *linear* birth-and-death process, the jumping chain is a simple random walk on $S = \{0, 1, 2, \dots\}$ with 0 as an absorbing state and (in the usual notation) $p = \lambda/(\lambda+\mu)$, $q = \mu/(\lambda+\mu)$.
Hence the probabilities of absorption in state 0, α_i^C, are just the ruin probabilities starting at state i and so

$$\alpha_i^{\{0\}} = \min\left(1,\ \left(\frac{q}{p}\right)^i\right) = \min\left(1,\ \left(\frac{\mu}{\lambda}\right)^i\right), \qquad i \geq 0.$$

6. (a) For $0 \leq t < t+h$ and $i = 1, 2, \dots, M-1$,

$$\begin{aligned} p_{0,1}(t, t+h) &= \lambda h + o(h) \\ p_{i,i-1}(t, t+h) &= i\mu h + o(h) \\ p_{i,i+1}(t, t+h) &= \lambda h + o(h) \\ p_{M,M-1}(t, t+h) &= M\mu h + o(h) \end{aligned}$$

and $p_{i,j}(t, t+h) = o(h)$ for all other combinations of i and j with $i \neq j$, since the probability of more than one machine breaking down or a machine breaking down and one being repaired in the short interval of length h is $o(h)$.
For example, $p_{i,i-2}(t, t+h)$ would be the probability that 2 machines break down or 3 machines break down and one is repaired, or other similar combinations in the short time interval h. But the probability that 2 machines break down is

$$(i\mu h + o(h))((i-1)\mu h + o(h)) = i(i-1)\mu^2 h^2 + o(h) = o(h)$$

while the probability of 3 machines breaking down and one being repaired is

$$(i\mu h+o(h))((i-1)\mu h+o(h))((i-2)\mu h+o(h))(\lambda h+o(h)) = i(i-1)(i-2)\mu^3\lambda h^4+o(h) = o(h)$$

as $h \to 0$. It therefore follows (with a bit of extra calculus regarding infinite series expansions) that $p_{i,i-2}(t, t+h) = o(h)$ as $h \to 0$.
Hence $Q = (q_{ij})$, where for $i = 1, 2, \dots, M-1$

$$\begin{aligned} q_{0,0} &= -\lambda \\ q_{0,1} &= \lambda \\ q_{M,M-1} &= M\mu \\ q_{M,M} &= -M\mu \\ q_{i,i+1} &= \lambda \\ q_{i,i-1} &= i\mu \\ q_{i,i} &= -(\lambda + i\mu) \end{aligned}$$

and $q_{ij} = 0$ for all other combinations of i and j.

(b) Note that since there are a finite number $(M+1)$ of states and since it is possible to go from any number of machines working to any other number of machines working, all states intercommunicate and so there exists a unique stationary distribution.

We first need to check that $\boldsymbol{\pi}$ as given satisfies $\boldsymbol{\pi}Q = \mathbf{0}$. The equation $\boldsymbol{\pi}Q = \mathbf{0}$ is equivalent to:

$$\begin{aligned} -\pi_0\lambda + \pi_1\mu &= 0 \\ \pi_{n-1}\lambda - \pi_n(\lambda + n\mu) + \pi_{n+1}(n+1)\mu &= 0, \qquad 0 < n < M \\ \pi_{M-1}\lambda - \pi_M M\mu &= 0. \end{aligned}$$

We now substitute the given π_n into the left hand sides of the above equations:

$$\begin{aligned} -k\frac{\left(\frac{\lambda}{\mu}\right)^0}{0!}\lambda + k\frac{\left(\frac{\lambda}{\mu}\right)^1}{1!}\mu &= -\lambda k + \lambda k = 0 \\ k\frac{\left(\frac{\lambda}{\mu}\right)^{n-1}}{(n-1)!}\lambda - k\frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}(\lambda + n\mu) + k\frac{\left(\frac{\lambda}{\mu}\right)^{n+1}}{(n+1)!}(n+1)\mu &= k\left(\frac{n\lambda^n\mu - n\lambda^n\mu - \lambda^{n+1} + \lambda^{n+1}}{\mu^n n!}\right) = 0 \\ l\frac{\left(\frac{\lambda}{\mu}\right)^{M-1}}{(M-1)!}\lambda - l\frac{\left(\frac{\lambda}{\mu}\right)^M}{M!}M\mu &= \frac{k\lambda^M k - k\lambda^M k}{(M-1)!\mu^{M-1}} = 0. \end{aligned}$$

For $\boldsymbol{\pi}$ to be a distribution we require $\sum_{n=0}^{M}\pi_n = 1$ and so

$$k = \left(\sum_{n=0}^{M}\frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n\right)^{-1}.$$

(c) We have

$$\begin{aligned} \lim_{M\to\infty}\pi_n &= \lim_{M\to\infty}\left(\sum_{n=0}^{M}\frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n\right)^{-1}\frac{\left(\frac{\lambda}{\mu}\right)^n}{n!} \\ &= \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}\left(\lim_{M\to\infty}\sum_{n=0}^{M}\frac{1}{n!}\left(\frac{\lambda}{\mu}\right)^n\right)^{-1} \\ &= \frac{\left(\frac{\lambda}{\mu}\right)^n}{n!}\exp\left(-\frac{\lambda}{\mu}\right). \end{aligned}$$

In other words, as $M \to \infty$, the distribution of the number of working machines converges to the Poisson distribution with mean λ/μ.